

Specification (Draft 1)

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1. INTRODUCTION

1.1 Context & Document Objective

The scout group wants a BI system for a global view, comparative analysis between seasons and units, and budget management, with an opening towards predictive capabilities.

This document translates these needs into clear specifications (data, KPIs, DW).

1.2 Identified Issues

<i>Problem</i>	<i>Impact</i>	<i>Available Data</i>
Data fragmented across 45+ Excel files	Lack of overall view, time-consuming manual analysis	Complete but scattered
Alarming drop in membership (-44% in 1 year)	Risk to the group's sustainability	3 seasons of historical data
Explosion of costs (+103% cost/member)	Severe budget constraints	Detailed budget/expense data
Activity load per member ×2.2 (paradox)	Risk of burnout for management staff	Activity vs. membership data
No anticipation of trends	Reactive vs. proactive planning	Sufficient time-based data
Lack of SDG indicators	Unable to measure social impact	Basic data available

2. Main Business Objectives

1. Centralize and structure all scattered scout data
 2. Improve the quality and speed of decision-making
 3. Understand past and current trends
 4. Anticipate future developments thanks to predictive analytics
 5. Introduce artificial intelligence mechanisms to support decision-makers
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3. Scope

1 : Members

Registrations and Renewals

- Recording new members
- Tracking seasonal renewals
- Standardized form with validation

Demographic Distribution

- View by unit (**6 units: Cubs, Brownies, Scouts, Guides, Rovers, Senior Guides**)
- Distribution by age
- Distribution by gender (M/F)
- History by season (2023-2024 to 2025-2026)

Membership Evolution

- Temporal trends (evolution charts)
- Growth/Decline rate
- Retention rate by cohort
"Indicator measuring the ability to retain members of the same group over time"
- Analysis of departure reasons (if data available)

2 : Scout Units

Unit Activity

- Number of activities organized per unit
- Participation rate per unit
- Specific calendar per unit
- Supervision tracking

Unit Performance

- Calculated performance indicators
- Comparative benchmarks between units
- Objective scores
- Historical progression

Unit Dynamics

- Evolution of unit composition
- Member turnover

- Average engagement
- Collaboration between units

3 : Activities

Weekly Meetings

- Planning and calendar
- Attendance/absence tracking
- Activity evaluation
- Resources used

Educational Outings

- Organization and logistics
- Participant list
- Detailed costs
- Educational feedback

Camps (Spring, Summer, Winter)

- Detailed planning
- Participant management
- Specific budget
- Post-camp evaluation

Special Events

- Participation in regional/national events
- Special projects (environment, social)
- External partnerships
- Measured impact

4 : Budgets

Budget Forecasts

- Annual budget per unit
- Budget per activity type
- Seasonal budget (summer/winter)
- Future needs predictions

Actual Expenses

- Recording expenses
- Categorization (transport ,food , equipment , others)
- Digitized receipts
- Real-time tracking

Resource Consumption

- Budget utilization rate
- Overspending alerts
- Optimization suggestions
- Calculated ROI

Financial Analysis

- Average cost per member
- Cost per activity type
- Budget vs. actual variances
- Spending trends

5: Temporal Analysis

Comparison by Scout Season

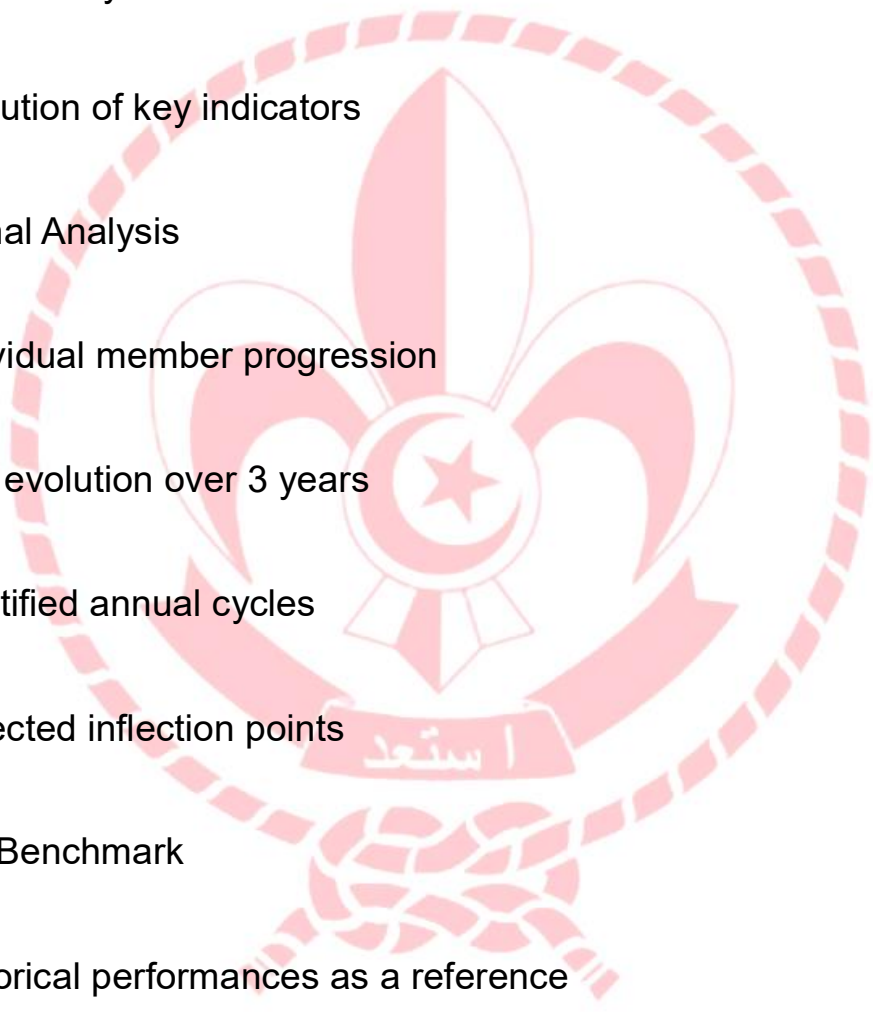
- Season 2023-2024 vs 2024-2025 vs 2025-2026
- Multi-year trends
- Seasonality of activities
- Evolution of key indicators

Longitudinal Analysis

- Individual member progression
- Unit evolution over 3 years
- Identified annual cycles
- Detected inflection points

Temporal Benchmark

- Historical performances as a reference
- Objectives based on history
- Measured improvements
- Identified best practices



6 : Predictive and Intelligent Analysis

Quantitative Predictions

- Future membership per unit
- Budget needed for next season
- Anticipated participation rate
- Human resource needs

Intelligent Classification

- Units classified by performance (A/B/C)
- Members segmented by engagement level
- Activities categorized by ROI
- Prioritized risks

Pattern Detection

- Operational anomalies
- Early warning signals for problems
- Improvement opportunities
- Emerging trends

Automated Recommendations

- Priority actions per unit
- Optimal resource allocation
- Planning suggestions
- Preventive alerts

4. STRATEGIC OBJECTIVES

4.1 Business Intelligence Objectives

	Objective	Description	Priority
BI-01	Structured member follow-up	View by unit, age,gender, season	● High

BI-02	Activity analysis	Completed vs. planned + participation	● High
BI-03	Budget tracking	Expenses associated with activities	● High
BI-04	Performance comparison	Between units and between seasons	□ Medium
BI-05	Identification of strengths/weaknesses	Areas for improvement	□ Medium

4.2 Machine Learning Objectives

Objective		Description
ML-01	Predict future membership	Per unit and season
ML-02	Estimate required budgets	Based on planned activity volume
ML-03	Predict participation rate	For activities
ML-04	Unit performance classification	(Low, medium, high)
ML-05	Identify at-risk units	Declining membership or disengagement
ML-06	Unit segmentation	Based on behavior and dynamics
ML-07	Member grouping	By engagement level
ML-08	Detection of atypical profiles	Anomalies revealing issues

4.3 Advanced Objectives -- Artificial Intelligence

Objective		Description
IA-01	Decision support system	Based on analytical results
IA-02	Automatic recommendations	Priority actions, resource allocation
IA-03	Weak signal detection	Organizational or financial risks
IA-04	Forward-looking scenarios	Evaluation of strategic choices

4.4 PREDICTIVE OBJECTIVES

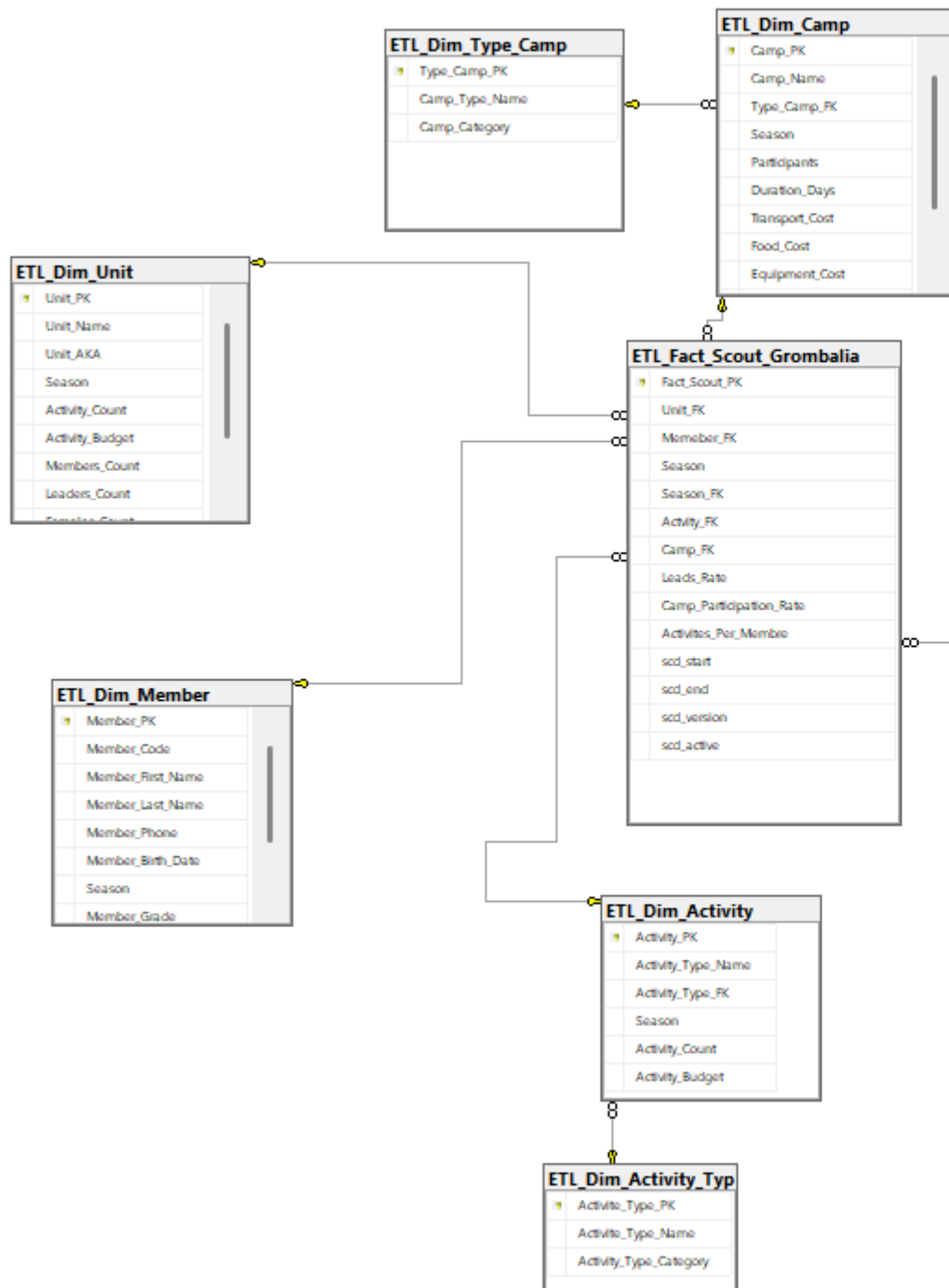
Predictive Objective	Description	Required Data
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P-01	Predict future membership	Forecast number of members per unit for next season	Membership history for 3 seasons
P-02	Estimate required budgets	Calculate financial needs based on planned activities	Historical costs + planning
P-03	Predict participation rate	Anticipate average participation in activities	Participation history
P-04	Forecast human resource needs	Estimate number of leaders needed per unit	Historical leader/member ratio

4.5 Sustainable Development Goals (SDGs)

SDG	Scout Objective	Contribution
SDG 4	<i>Quality Education</i>	<i>Scout educational programs</i>
SDG 5	<i>Gender Equality</i>	<i>Gender parity in units</i>
SDG 10	<i>Reduced Inequalities</i>	<i>Inclusion of all socio-economic profiles</i>
SDG 11	<i>Sustainable Cities</i>	<i>Local community engagement</i>
SDG 12	<i>Responsible Consumption</i>	<i>Sustainable resource management</i>
SDG 16	<i>Effective Institutions</i>	<i>Governance transparency</i>
SDG 17	<i>Partnerships</i>	<i>Multi-actor collaborations</i>

5) DW Modeling



5) KPIs (calculated on F AllMeasures)

Members :

- Membership = SUM(NbRegistered). Critical. => This represents the total number of registered members.

- **Membership evolution => Ça mesure la croissance ou la baisse du nombre d'adhérents par rapport à la saison précédente.**
- **Activity/Camp participation = $\text{ParticipantsCamp} / \text{NbRegistered}$ (or other proxy). Critical. => This measures the participation rate in camps or – activities.**
- **Average number of activities / unit = $\text{SUM}(\text{NbActivities}) / \text{COUNT}(\text{D_Unit})$. Important. => Average number of activities organised per unit.**
- **Total budget per season = $\text{SUM}(\text{BudgetUnit} + \text{BudgetActivity} + \text{BudgetCamp})$. Critical. => budget global of the season**
- **Average cost per member = $\text{Total budget} / \text{NbRegistered}$. Critical. => How much does a member cost on average?**
- **At-risk units (simple rules): drop in Registered $\geq X\%$ or low NbActivities or Leaders/Registered ratio $<$ threshold. Critical.**

Budgets :

- **Budget total**
- **Budget moyen par unité = $\text{TotalBudget} / \text{Nombred'unites}$**
- **Coût moyen par adhérent**

Media :

- **Total posts (season) => Total number of social media posts published by Grombalia**
- **Posts per month => Normalized activity indicator**
- **Average engagement per post => $(\text{likes} + \text{comments} + \text{shares}) / \text{posts}$**
- **Active platforms => Number of platforms where Grombalia posted during the season**
- **Relative visibility index => $\text{Grombalia posts} \div \text{benchmark average posts}$**

6) Project decision makers

1. **Unit heads**
 2. **Finance managers**
 3. **Scout Group Leader**
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7) Quality & Risks

- **Label differences (FR/AR), invalid dates (" 8 2025 نوفمبر 9"), Total rows --- to correct.**
- **Short history for some series (impact on prediction). These points are highlighted in the document (need for reliable and comparable data).**

